01 / UNEQUAL LINK STUDY

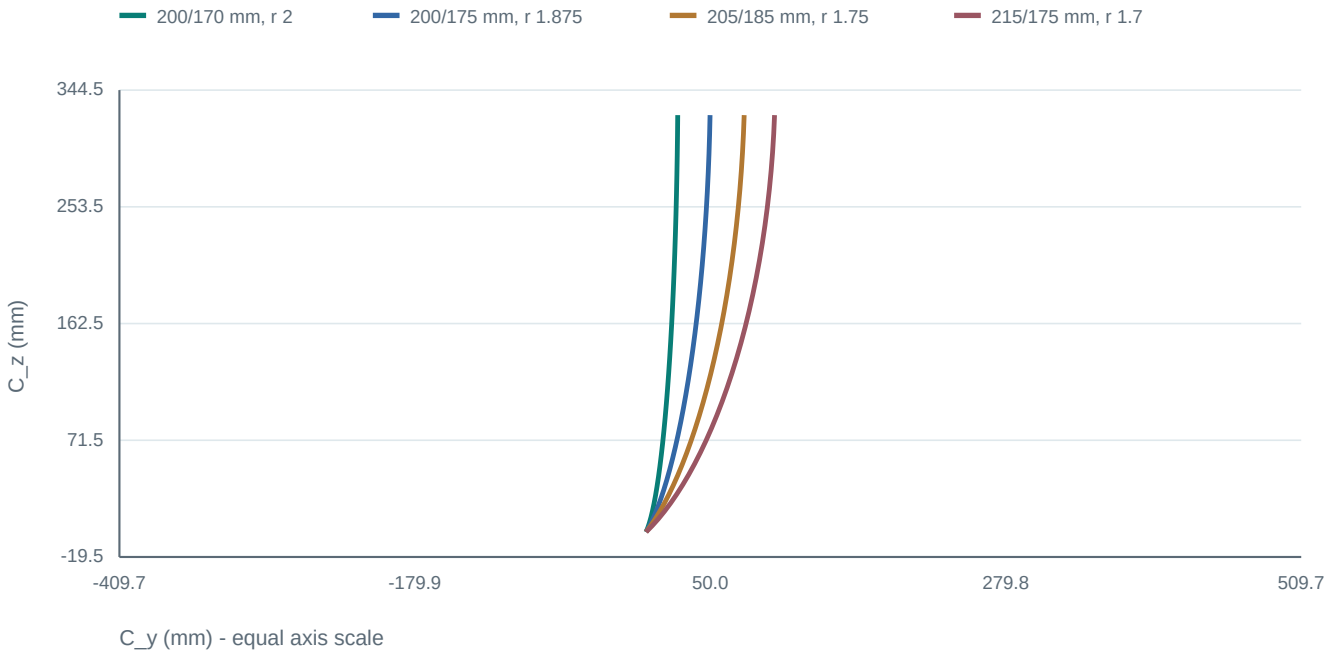
Upper-long links within 1.25x

Owner correction: longer /shorter <=1.25; alpha=phase=0; wheel R200 mm.

Yes, the upper link can be longer. These four rounded examples obey the new length constraint and provide **325 mm total vertical travel, including assumed 20+20 mm reserves: 285 mm useful.** Their ideal height maps stay invertible through the fold. The earlier 160/475 mm proposal is outside this constraint and is superseded.

Upper /lower mm	Length ratio	Guide r	Sideways mm	Hip peak Nm
200 /170	1.176	2	24.62	40.30
200 /175	1.143	1.875	49.77	38.52
205 /185	1.108	1.75	76.31	37.50
215 /175	1.229	1.7	99.95	36.89

Normal paths aligned at their start; equal physical axis scales



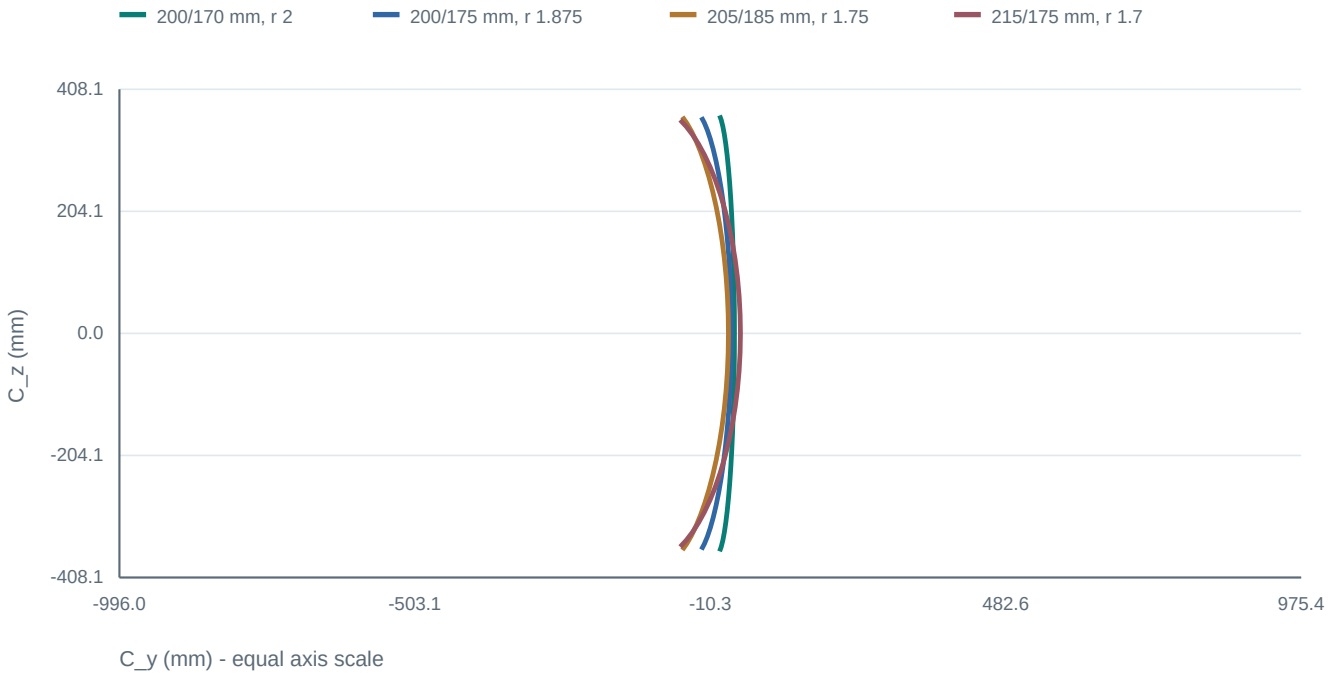
Balanced study seed: 200/175 mm, guide 15:8 (1.875). This gives 49.77 mm sideways movement with a 1.143 length ratio. The 205/185 mm, 7:4 alternative gives 76.31 mm and slightly lower static torque. All require axial wheel/link separation at the fold; none is a collision-free coplanar mechanism.

02 / UNEQUAL LINK STUDY

Curved paths through the fold

Full ideal q=100-260 deg; the 325 mm normal stroke is only part of this envelope.

Full wheel-centre paths in hip coordinates; equal physical scales



Upper /lower mm	Full height mm	Min Jz mm/rad	Useful hip ROM deg
200 /170	728.8	64.25	59.32
200 /175	722.8	87.10	60.83
205 /185	724.2	104.97	61.46
215 /175	713.6	105.84	62.20

With a=L2,b=L3,t=q-pi,k=r-1: `x=a cos(t)-b cos(kt); z=a sin(t)+b sin(kt)`. Jz is positive over the full envelope, so each height has one q through 180 degrees. The fold tangent is vertical; full endpoint sideways drift is zero by symmetry, despite nonzero lateral excursion.

At r=2 with unequal links, x=(a-b)cos(t), z=(a+b)sin(t): an ellipse. A 2:1 guide therefore remains a useful curved-path option. It becomes a vertical line only when a=b. Four labelled six-second offline GIFs below show actual wheel radius and the projected overlaps; they do not simulate rolling or collision.

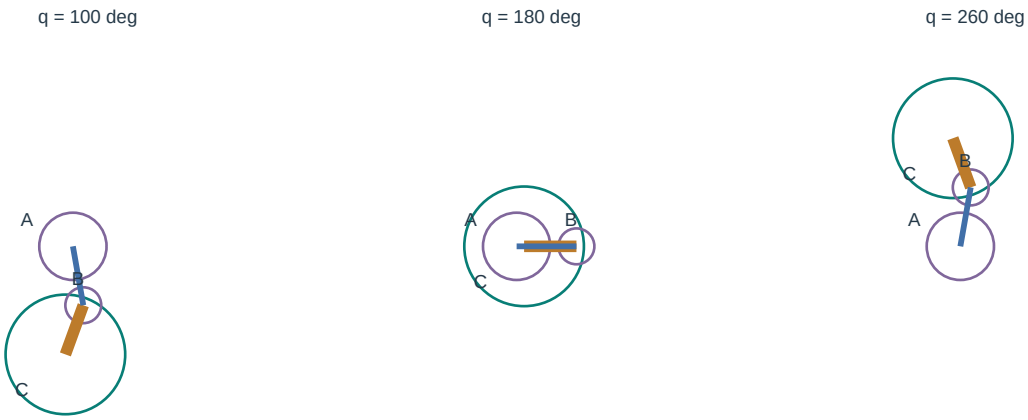
03 / UNEQUAL LINK STUDY

Axial separation is necessary

Near-equal upper-long paths invert ideally; projected wheel and rod interference remains.

At the fold, B=(a,0), C=(a-b,0). For a>b, **0<a-b<a**, so the wheel centre lies **inside the upper segment AB**. Its projected distance to that segment is zero at any scale. A flat coplanar wheel/upper-link layout cannot pass through this fold, even if hip-centre clearance alone looks acceptable.

200/175 mm, guide 15:8: wheel and guide pitch circles; blue upper /orange lower



Upper /lower mm	Fold hip distance mm	Wheel /upper gap mm	Guide pitch surplus mm
200 /170	30.0	-200.0	20.0
200 /175	25.0	-200.0	27.5
205 /185	20.0	-200.0	40.0
215 /175	40.0	-200.0	53.0

A negative wheel/upper gap means projected overlap. All rows pass the assumed 10 mm guide pitch-centre surplus a-60(r+1), with rb 60 and ra=60r; that guide check does not establish wheel, flange, shaft, belt or chassis clearance.

For a lower-long alternative, minimum wheel/hip and wheel/thin-upper-segment distance is b-a. Under the 1.25x limit, merely reaching 220 mm requires shorter link>=880 mm and longer>=1100 mm. Adding hip pitch radius requires still more. That is outside this compact 150-400 mm search. No searched candidate passes the full coplanar wheel clearance screen.

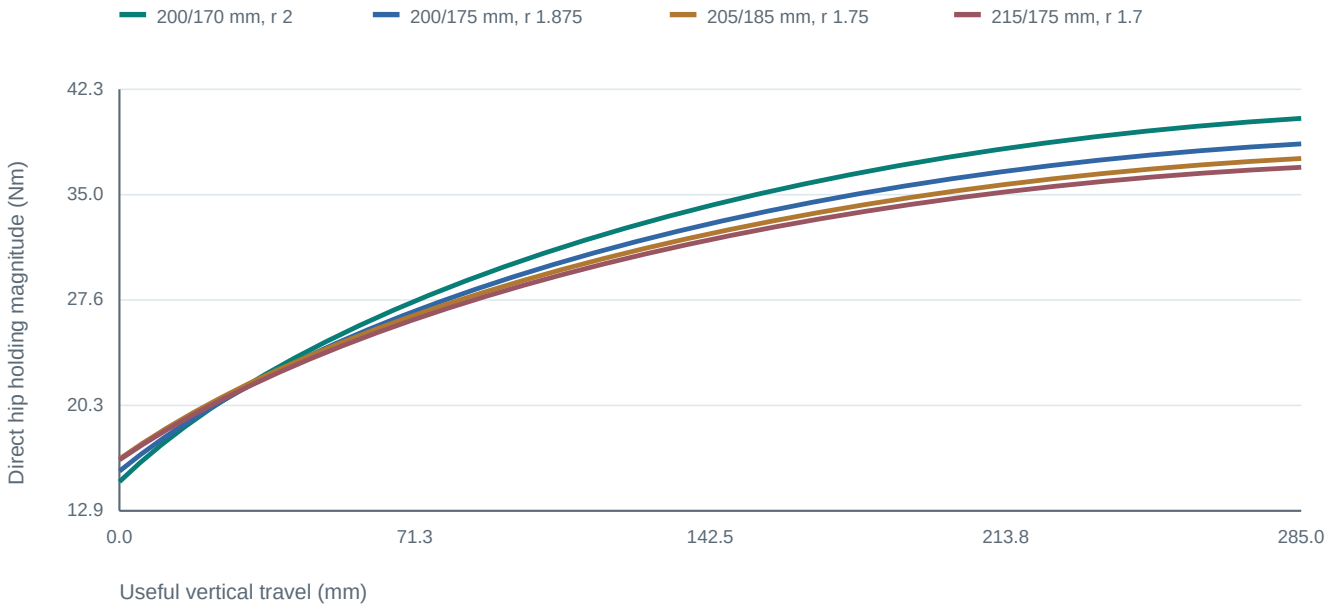
Proposed packaging direction: place wheel and links in separate axial planes, using an appropriate fork/offset/shaft arrangement. Rods also overlap in projection at the fold. Actual axial stack, tire width, chassis and outer hardware must be measured in CAD; physical full inversion remains unvalidated.

04 / UNEQUAL LINK STUDY

Force and control tradeoffs

Pure vertical equivalent corner load 110.3625 N =45 kg x9.81/4; no spring or moving-link inertia.

Individual direct-hip holding torque magnitudes over 285 mm useful travel



Upper /lower mm	Peak /RMS Nm	Hip speed rad/s	Side-load peak Nm
200 /170	40.30 /32.99	1.848	40.46
200 /175	38.52 /31.82	1.761	38.87
205 /185	37.50 /31.20	1.670	38.06
215 /175	36.89 /30.78	1.677	37.54

Holding magnitude is Fz Jz for the upward-load case; the signed balancing hip torque is -Fz Jz. Reducing maximum Jz improves worst vertical force per hip Nm. A smaller Jz at a given pose needs greater joint speed for the same vertical speed; peak speed depends on minimum Jz. Speed assumes constant vertical 0.25 m/s; side-load peak is a both-sign |Fx/Fz|<=0.3 assumed sensitivity.

Relative-knee torque is **-tau_hip/r**; relative-knee speed is **-r times hip speed**. Half-torque applies only at r2. Motor gearing is another mapping. Height-RMS equals time-weighted torque RMS at constant vertical speed without dwell; heating still needs motor/gear losses and actual duty.

Ideal useful lift work remains 31.453 J per leg in every case. Lower torque is mechanical advantage, not free energy. Total supported loads 3g and 4g multiply the static load term by 3 and 4; they do not predict impacts. Permitted sideways movement is not measured compliance: stiffness, contact, rolling and inertia are unresolved. No new spring or motor rating is selected.

05 / UNEQUAL LINK STUDY

Hard length limit, separate objectives

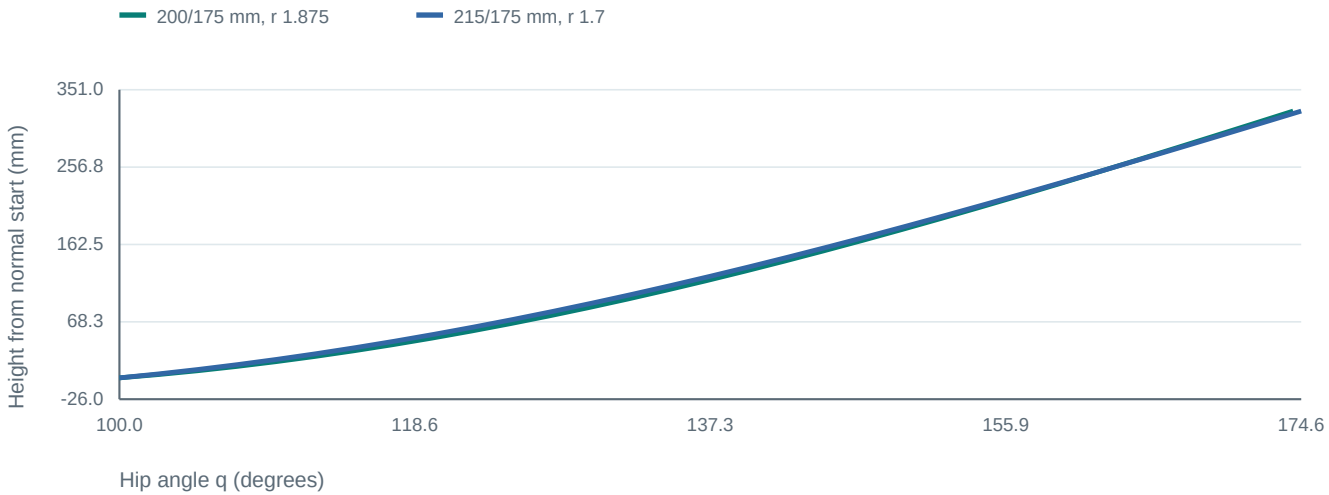
6,601,338 bounded geometry/window combinations; no weighted score or R2 method.

L2 and L3=150-400 mm in 5 mm steps; max/min<=1.25 and |L2-L3|>=5 mm. Guide 1.2-2.6 in 0.005 steps plus exact 5/3. Normal q start 100-140 in 5 degree steps; end solves 325 mm height and must be<=175. Full 100-260 requires Jz>=5 mm/rad. Sideways excursion 25/50/75/100 +/-2.5 mm, without reversal.

Upper-long and lower-long families are ranked separately. Hard constraints come first; minimize useful peak Jz and total link sum as separate objectives. Coarse 33 poses retain eight candidates per bucket; finalists get continuous derivative-extrema checks and 2001-pose integrals. Finite grid and retained pool do not prove a global or manufacturing optimum.

50 mm comparison	Upper /lower, r	Hip peak Nm	Length ratio
Upper-long force leader	190/185, 1.815	37.26	1.027
Rounded 15:8 example	200/175, 1.875	38.52	1.143
Upper-long size leader	200/170, 1.88	38.19	1.176
Lower-long force leader	180/210, 1.715	36.13	1.167

Height reserves trim the angular control window



Useful ROM uses height roots z^-1(zstart+20) and z^-1(zend-20). For 200/175,15:8, total q=100-174.083 deg; useful hip ROM 60.829 deg. Do not subtract 20 degrees for 20 mm. Rounded ratios balance readable parameters, compactness and lateral target; their somewhat higher torque than the grid force leaders is reported explicitly.

06 / UNEQUAL LINK STUDY

Checks and next design decision

Ideal inverse verified; collision-free axial packaging and loaded dynamics remain open.

23 numerical tests pass, including seven unequal-link tests. 7,440 original Jonathan V2/Andy 2D scalar comparisons across 20 cases agree, maximum 1.705e-13 mm/Nm. The 1.25x constraint and upper-long fold overlap have explicit regression checks. Independent read-only review confirms the four rounded examples and ellipse relation.

Actual Pymunk case	2 ms mm	1 ms mm	0.5 ms mm	0.25 ms mm
upper25	0.08461	0.04200	0.02093	0.01044
upper50	0.07441	0.03694	0.01840	0.00919
upper100	0.07279	0.03614	0.01801	0.00899

Actual positive-mass motor/pivot/gear constraints generate the three angular sweeps; only initial poses are assigned. The compact evidence retains every timestep and unchanged tolerance result, including coarse failures if present. Zero gravity/forces and no collision shapes mean geometry evidence, not validated loaded torque, contact, compliance, impact or physical inversion.

Start CAD with 200/175 mm, guide 15:8 for approximately 50 mm sideways movement, or 205/185 mm, 7:4 for about 75 mm. Both fit the required length relation. Their normal strokes approach the assumed 175 degree guard; inspect actual stop and buffer tolerances rather than treating these limits as measured hardware.

The earlier broad search recommended 160/475 mm before the owner clarified the 1.25x limit. That recommendation is superseded; its historical evidence remains in the preceding local revision and contribution log. This report now applies the constraint to every retained case. Existing Motion Lab models, earlier reports/spring cases and historical dual-rope **IMPOSSIBLE /rejected** stay intact.

Reproduction commands, source hashes, exact assumptions and validation history are in docs/unequal-inversion-study.md and docs/validation.md. The manual launcher opens this static file with adjacent GIFs; no server or new Motion Lab control is needed. Source base aaed324; workspace scope; canonical physics NOT APPLICABLE.